

1

RM-Class-11-12



- Class-11-12: Univariate and Bivariate Analysis of Data (Covers chapter-11)

2

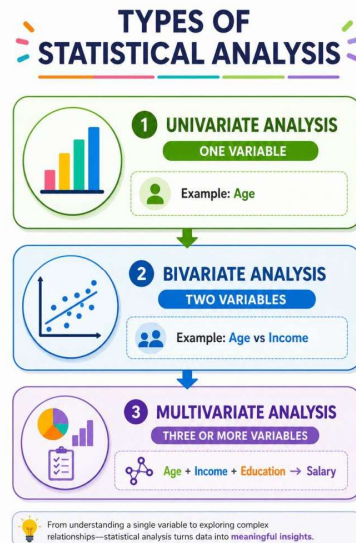
Learning Objective

By the end of this class, a student must be able to answer these questions.

- Distinguish between univariate, bivariate and multivariate analysis of Data
- Differentiate between the descriptive and inferential analysis based on the statistical approaches
- How descriptive analysis are applied to univariate data
- How descriptive analysis are applied to bivariate data
- What is rank order and data transformation in data analysis

3

Distinguish between univariate, bivariate and multivariate analysis of Data



4

Univariate analysis of data

- Univariate analysis means **analysing only one variable at a time**.
- Example:** Suppose a researcher asks 100 students: "How many hours do you study per day?"
- Here, only one variable is being studied: **hours of study**.
 - "Uni" means one.** It studies one characteristic or variable at a time.
 - Its main purpose is usually to describe or summarize data.
 - Common tools include mean, median, mode, percentages, frequency tables, range and standard deviation.
 - Bar charts, pie charts and histograms may also be used to present univariate data.
 - It does not normally examine relationships between variables.

Bivariate analysis of data

5

- Bivariate analysis means **analysing two variables together to find whether there is any relationship or association between them.**

- Simple example:** Suppose a researcher studies:

- Variable 1: **Hours of study**
- Variable 2: **Marks obtained**

The researcher wants to know: "Do students who study for more hours obtain higher marks?"

Now two variables are being studied together. Therefore, it is **bivariate analysis**.

- Important points for students:**

- "Bi" means **two**. It examines **two variables at the same time**.
- Its main purpose is often to determine whether the variables are **related, associated or different from each other**.
- Cross-tabulation** is a common method for examining two categorical variables.
- Other techniques may include **correlation, simple regression, t-tests and chi-square tests**, depending on the type and measurement level of the variables.
- A relationship between two variables does not automatically prove that **one causes the other**.

Multivariate analysis of data

6

- Multivariate analysis means **analysing more than two variables at the same time**.

- Example:** Suppose a researcher wants to understand students' academic performance using:

- Hours of study
- Attendance
- Family income
- Use of social media
- Marks obtained

Here, several variables are being considered together. Therefore, it is **multivariate analysis**.

- Example, the researcher may ask: "How do study hours, attendance and social-media usage together influence students' marks?". This requires multivariate analysis.
- "Multi" means **many**. It involves **more than two variables**.
- It helps researchers study **complex situations** where several factors operate together.
- It can help identify the relative importance of different variables.
- Common multivariate techniques in business research include **multiple regression, factor analysis, cluster analysis, discriminant analysis and conjoint analysis**.

Comparison

7

Type of analysis	Number of variables	Main question	Simple example
Univariate	1	What does this variable look like?	What is the average salary of employees?
Bivariate	2	Are these two variables related?	Is salary related to work experience?
Multivariate	More than 2	How do several variables work together?	How do experience, education and skills together influence salary?

Univariate → Describe **ONE** variable
 Bivariate → Study the relationship between **TWO** variables
 Multivariate → Study **MANY** variables together

8

Differentiate between the descriptive and inferential analysis based on the statistical approaches

The Two Main Types of Statistical Analysis

Descriptive Statistics



Summarize and describe data (e.g., average age or salary). Show what happened but don't predict.

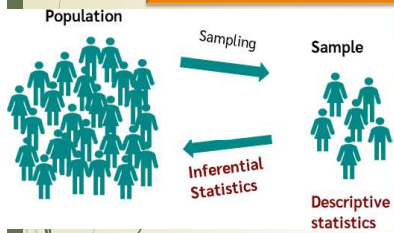
Inferential Statistics



Use a sample to make conclusions or predictions about a larger group (e.g., survey 200 customers to represent all).

9

Statistics in Research Methodology

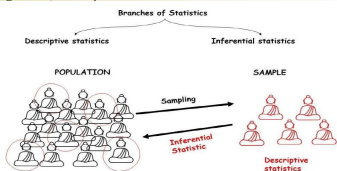


- We can broadly divide statistical techniques in research methodology into two major categories:

- **Descriptive Analysis (Descriptive Statistics)**

- **Inferential Analysis (Inferential Statistics)**

- The classification is based on the purpose of statistical analysis.



10

Descriptive Analysis

- Descriptive analysis refers to the statistical techniques **used to summarize, organize, and present the characteristics of collected data without making predictions or generalizations beyond the data.**
- Example: A researcher collects data on the monthly income of 200 employees and calculates:
 - Average income
 - Highest and lowest income
 - Percentage of employees in different income groups
 - The researcher is only describing the data collected.

11

Continue...

- It describes **what the data shows.**
- It does not test hypotheses or make conclusions about a larger population.
- It is usually the first step in data analysis.
- It includes:
 - Frequency distribution
 - Percentage analysis
 - Mean, median, mode
 - Range, variance, standard deviation
 - Tables, charts, graphs
- Examples of research questions:
 - What is the average age of respondents?
 - What percentage of customers prefer a particular brand?
 - What is the distribution of students' marks?

12

Inferential Analysis

- Inferential analysis refers to statistical techniques used to **draw conclusions, make predictions, or test hypotheses about a population based on data collected from a sample.**
- **Example:** A researcher surveys 500 voters from a city and uses the results to predict the voting preference of the entire city population.
- Here, the researcher is **making an inference from a sample to a population.**

Continue...

13

- It goes beyond describing the collected data.
- It helps researchers make decisions and test theories.
- It involves estimating population characteristics and testing relationships.
- It includes:
 - Hypothesis testing
 - t-test
 - z-test
 - Chi-square test
 - ANOVA
 - Correlation analysis
 - Regression analysis
- Examples of research questions:
 - Is there a significant relationship between advertising expenditure and sales?
 - Does training improve employee performance?
 - Are two groups significantly different from each other?

Relationship Between the Two Classifications

14

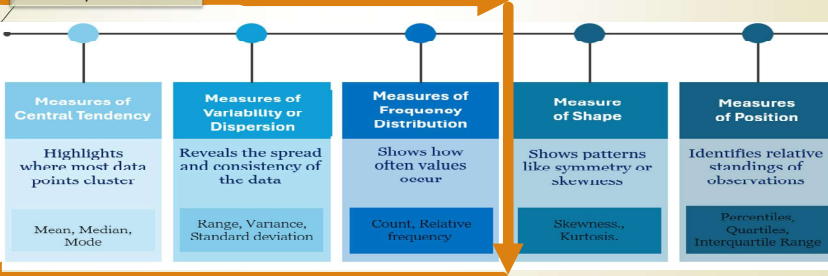
- The earlier classification of **univariate, bivariate, and multivariate analysis** is based on the **number of variables involved**, whereas **descriptive and inferential analysis** is based on the **purpose of analysis**.

Broad Division	Type of Analysis	Example
Descriptive Analysis	Univariate	Average age of respondents
Descriptive Analysis	Bivariate	Cross-tabulation of gender and brand preference
Inferential Analysis	Bivariate	Testing relationship between income and purchase behaviour
Inferential Analysis	Multivariate	Multiple regression to predict sales using several factors

15

How descriptive analysis are applied to univariate data

These portions are within syllabus

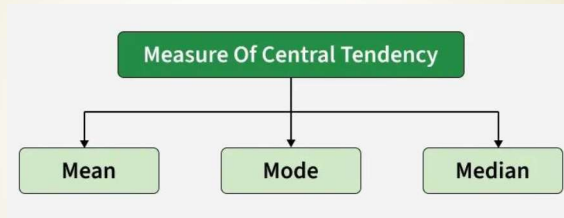


16

Importance of teaching descriptive statistics

Teaching basic descriptive statistics is essential in a research methodology class because it forms the foundation for understanding data before applying advanced statistical methods.

17



18

1: Measure of central tendency

These describe the centre of a data set. In other words, it gives measures of central tendency that **give you the average for each response.**

Mean (Arithmetic Average)

Median (Middle value)

Mode (Most frequent value)

19

1(a): Mean

- Mean, also known as **arithmetic average**.
- It is the most common measure of central tendency and may be defined as the value we get by dividing the total **values of various given items in a series by the total number of items.**

$$\text{Mean (or } \bar{X}) = \frac{\sum X_i}{n} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

where $\bar{X}$ = The symbol we use for mean (pronounced as X bar)

$\sum$ = Symbol for summation

X_i = Value of the i th item X , $i = 1, 2, \dots, n$

n = total number of items

The Arithmetic Mean

$$\text{The mean} = \frac{\text{Sum of values}}{\text{Number of values}} = \frac{6 + 5 + 9 + 1 + 2}{5}$$

$$\text{The mean} = \frac{20}{5} = 4$$

20

Geometric mean

- It is defined as the **n th root of the product of the values of n times in a given series.**

$$\begin{aligned} \text{Geometric mean (or G.M.)} &= \sqrt[n]{\pi X_i} \\ &= \sqrt[n]{X_1 \cdot X_2 \cdot X_3 \dots X_n} \end{aligned}$$

where

G.M. = geometric mean,

n = number of items.

X_i = i th value of the variable X

π = conventional product notation

For instance, the geometric mean of the numbers, 4, 6, and 9 is worked out as

$$\begin{aligned} \text{G.M.} &= \sqrt[3]{4 \cdot 6 \cdot 9} \\ &= 6 \end{aligned}$$

21

Harmonic mean

- **Harmonic mean** is defined as the reciprocal of the average of reciprocals of the values of items of a series.

$$\begin{aligned}\text{Harmonic mean (H.M.)} &= \text{Rec.} \frac{\sum \text{Rec.} X_i}{n} \\ &= \text{Rec.} \frac{\text{Rec.} X_1 + \text{Rec.} X_2 + \dots + \text{Rec.} X_n}{n}\end{aligned}$$

where

H.M. = Harmonic mean
Rec. = Reciprocal
 X_i = i th value of the variable X
 n = number of items

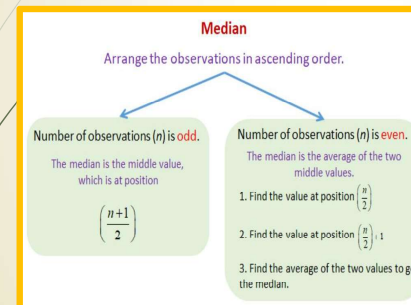
For instance, the harmonic mean of the numbers 4, 5, and 10 is worked out as

$$\begin{aligned}\text{H.M.} &= \text{Rec.} \frac{1/4 + 1/5 + 1/10}{3} = \text{Rec.} \frac{15 + 12 + 6}{60} \\ &= \text{Rec.} \left(\frac{33}{60} \times \frac{1}{3} \right) = \frac{60}{11} = 5.45\end{aligned}$$

22

1 (b): Median

- The median is **the middle number in a sorted, ascending or descending list of numbers.**



1, 3, 3, **6**, 7, 8, 9

Median = **6**

1, 2, 3, **4**, **5**, 6, 8, 9

Median = $(4 + 5) \div 2$
= **4.5**

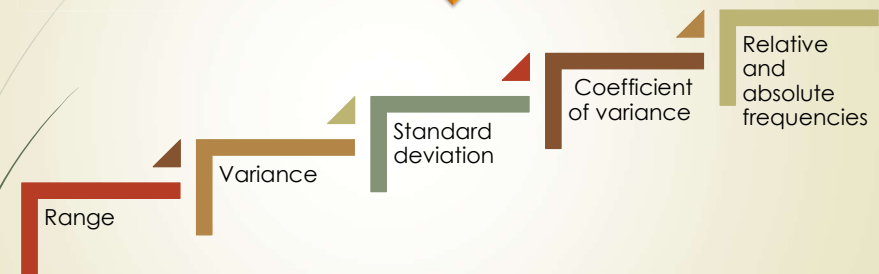
23

1 (c): Mode

- Mode is the most commonly or **frequently occurring value in a series.**
- A mode is defined as the value that has the highest **frequency** in a given set of values. It is the value that **appears the most frequently.**
- A dataset may not have any mode or there may be more than one mode in a dataset.
- **Example:** In the given set of data: 2, 4, 5, 5, 6, 7, the mode of the data set is five since it has appeared in the set twice.
- **Example:** Data Series: 4, 7, 4, 5, 9, 4, 7, 7, 6. In this dataset, both 4 and 7 have the highest frequency, each appearing three times. Therefore, the dataset is bimodal (it has two modes: 4 and 7).

24

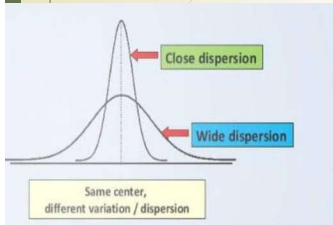
Measures of dispersion



2. Measures of dispersion

25

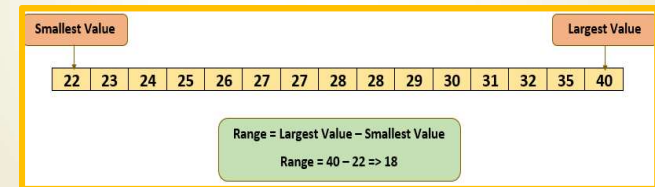
- Dispersion **is the state of getting dispersed or spread.**
- Statistical dispersion means **the extent to which numerical data is likely to vary about an average value.**
- Dispersion **helps to understand the distribution of the data.**
- In statistics, the measures of dispersion help to **interpret the variability of data** i.e. to know how much homogenous or heterogeneous the data is. In simple terms, it shows how squeezed or scattered the variable.



2(a): Range

26

- It is the simplest possible measure of dispersion and is defined as the **difference between the values of the extreme items of a series.**
- Range** = (Highest value of an item in a series) - (Lowest value of an item in a series).
- Example: 1, 3, 5, 6, 7 $\Rightarrow$ Range = 7 - 1 = 6



27

2 (b) Variance

- Variance is a **measure of how data points differ from the mean.** According to Layman, a variance is a measure of how far a set of data (numbers) are spread out from their mean (average) value.
- Variance means **to find the expected difference of deviation from the actual value.**



Population

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

μ - Population Average
 x_i - Individual Population Value
 n - Total Number of Population
 σ^2 - Variance of Population

Sample

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$\bar{x}$ - Sample Average
 x_i - Individual Population Value
 n - Total Number of Sample
 S^2 - Variance of Sample



Calculate the population variance of the following data set:

1, 2, 5, 7, 10

$N = 5$

x_i	$x_i - \mu$	$(x_i - \mu)^2$
1	-4	16
2	-3	9
5	0	0
7	2	4
10	5	25
Total:		54

$$\mu = \frac{1 + 2 + 5 + 7 + 10}{5}$$

$$\mu = 5$$

Calculate the Population Variance

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$54 \div 5 = 10.8$$



Calculate the sample variance of the following data set:

3, 4, 6, 7, 7, 9, 13

$n = 7$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
3	-4	16
4	-3	9
6	-1	1
7	0	0
7	0	0
9	2	4
13	6	36
Total:		66

Calculate the Sample Variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$66 \div 6 = 11$$

The variance = 11

31

2(c): Standard deviation

- Standard deviation is defined as the **square root of the average of the squares of deviations**, when such deviations for the values of individual items in a series are obtained from the arithmetic average.
- In statistics, the standard **deviation is a measure of the amount of variation or dispersion of a set of values**. A low standard deviation indicates that the values tend to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range.

Standard Deviation Formula

$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

X – The Value in the data distribution
 μ – The population Mean
 N – Total Number of Observations

$$s = \sqrt{\frac{\sum (X - \bar{x})^2}{n-1}}$$

X – The Value in the data distribution
 $\bar{x}$ – The Sample Mean
 n – Total Number of Observations

Population Standard Deviation & Variance

The math test scores of all students in a small study group are: 88, 74, 85, 78, 90. Find the population variance and standard deviation.

x_i	μ	$x_i - \mu$	$(x_i - \mu)^2$
88	83	5	25
74	83	-9	81
85	83	2	4
78	83	-5	25
90	83	7	49

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\sigma = \sqrt{\sigma^2}$$

Calculation Example: Sample Standard Deviation

Sample
Data (X_i):

10 12 14 15 17 18 18 24

$n = 8$ Mean = $\bar{X} = 16$

$$S = \sqrt{\frac{(10 - \bar{X})^2 + (12 - \bar{X})^2 + (14 - \bar{X})^2 + \dots + (24 - \bar{X})^2}{n - 1}}$$

$$= \sqrt{\frac{(10 - 16)^2 + (12 - 16)^2 + (14 - 16)^2 + \dots + (24 - 16)^2}{7}}$$

$$= \sqrt{\frac{130}{7}} = 4.308 \rightarrow \text{A measure of the "average" scatter around the mean}$$

Compute the standard deviation of the following sample:
2, 3, 6, 6, 8

x	$x - \bar{x}$	$(x - \bar{x})^2$
2	2-5=-3	$(-3)^2=9$
3	3-5=-2	$(-2)^2=4$
6	6-5=1	$1^2=1$
6	6-5=1	$1^2=1$
8	8-5=3	$3^2=9$

$$\bar{x} = \frac{25}{5} = 5$$

$$SS=24$$

Step 1: Compute the sample mean $s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$

Step 2: Subtract the mean from each value (i.e., compute the deviations)

Step 3: Square each deviation

Step 4: Add the deviations to get the sum of squares (SS)

Step 5: Divide the SS by n-1 to get the variance (s^2)

Step 6: Take the square root of the variance to get the standard deviation

$$s^2 = \frac{24}{5 - 1} = \frac{24}{4} = 6$$

$$s = \sqrt{6} = 2.449$$

35

2(d) Coefficient of variance

- The **Coefficient of Variation (C.V.)** is a **relative measure of dispersion**. It is used to compare the variability or consistency of two or more datasets, especially when their means are different.

The formula is:

$$C.V. = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

Where:

- Standard Deviation (σ) = measure of dispersion
- Mean ($\bar{X}$) = average of the observations

Interpretation

The lower the coefficient of variation, the greater the consistency or stability of the data.

The higher the coefficient of variation, the greater the variability and the less consistent the data.

36

Example: Coefficient of Variance

- Suppose two researchers have the following results:

Researcher	Mean	Standard Deviation	C.V.
A	50	5	10%
B	60	9	15%

For A:

$$C.V. = \frac{5}{50} \times 100 = 10\%$$

For B:

$$C.V. = \frac{9}{60} \times 100 = 15\%$$

Therefore, Researcher A's data is more consistent, because its C.V. (10%) is lower than B's (15%).

37

2(e) Relative and absolute frequencies

- Frequency means the **number of times a particular value or observation occurs** in a dataset.
- For example, suppose the data is: 2, 3, 2, 5, 3, 2, 4, 5, 3

Measure of
frequency
variation

Value	Frequency
2	3
3	3
4	1
5	2

38

Measure of frequency distribution: Relative Frequency

Relative Frequency

Definition

Frequency in mathematics is a measure of how often a quantity is present and represents the chances of occurrence of that quantity.

Formula

$$\text{Relative Frequency} = \frac{\text{Subgroup Count}}{\text{Total Count}} = \frac{f}{n}$$

RELATIVE FREQUENCY

Color	Frequency	Relative Frequency
Purple	7	7/20= 35%
Blue	3	3/20= 15%
Pink	5	5/20=25%
Orange	5	5/20=25%
Total	20	20/20 = 100%

Number of books	Frequency	Relative frequency
0	2	$\frac{2}{18} = 0.111 \dots$
1	3	$\frac{3}{18} = 0.167$
2	5	$\frac{5}{18} = 0.278$
3	4	$\frac{4}{18} = 0.222 \dots$
4	4	$\frac{4}{18} = 0.222 \dots$

39

Measure of frequency distribution: Cumulative frequency

- Cumulative Frequency** is the running total of frequencies **up to a certain point** in a frequency distribution.
- It helps answer questions like "How many values are **less than or equal to** a certain value?"

School Grade	Frequency of Students	Cumulative Frequency
1	23	23
2	20	23 + 20 = 43
3	15	43 + 15 = 58
4	12	58 + 12 = 70
5	10	70 + 10 = 80
6	8	80 + 8 = 88

Scores: 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 5

Score	Frequency	Cumulative Frequency
1	2	2
2	5	7
3	4	11
4	2	13
5	1	14

Cumulative Frequency for Score 3 is 2+5+4 = 11

Age group (Class)	Frequency	Cumulative frequency	% Cumulative frequency	Relative Frequency
20-24	2	2	4.0	0.04
25-29	3	5	10.0	0.06
30-34	13	18	36.0	0.26
35-39	18	36	72.0	0.36
40-44	8	44	88.0	0.16
45-49	4	48	96.0	0.08
50-54	2	50	100.0	0.04
Total	50			

How descriptive analysis are applied to bivariate data

Descriptive Analysis of Bivariate Data

Bivariate data means data involving two variables. The purpose is to examine whether and how the two variables are related.

For example:

- ▶ Education and income
- ▶ Age and income
- ▶ Gender and employment status
- ▶ Training received and job performance

1. Cross Tabulation

- ▶ **Cross tabulation** is a method of presenting the relationship between **two categorical variables** in the form of a table.
- ▶ One variable is shown in **rows** and the other in **columns**.
- ▶ Suppose a researcher wants to examine the relationship between **gender** and **employment status** among 100 respondents.

Gender	Employed	Unemployed	Total
Male	40	10	50
Female	25	25	50
Total	65	35	100

This is a **cross-tabulation** of gender and employment status.

2. Cross Tabulation Using Survey Data

- ▶ In research methodology, cross-tabulation becomes particularly useful when the researcher has collected information through a **survey questionnaire**.
- ▶ Suppose a survey asks respondents:
 - ▶ **Q1. Do you use the internet for educational purposes?**
 - ▶ **Q2. What is your level of education?**
- ▶ The researcher can cross-tabulate the answers.

Education	Yes	No	Total
School	20	30	50
Graduate	40	10	50
Total	60	40	100

Now the researcher can examine whether educational level is associated with the use of the internet for educational purposes.

3. Questions

45

- This part is important because **cross-tabulation is not simply about making a table**. The researcher should ask meaningful questions about the relationship between the two variables.

- For example, suppose we have:

Education	Satisfied	Not satisfied
Graduate	45	5
Non-graduate	25	25

Instead of merely presenting the table, the researcher asks:

Is satisfaction related to educational level?

Or:

Are graduates more likely to be satisfied than non-graduates?

The table helps the researcher answer such questions.

4. Elaboration of Cross Table

46

- Elaboration of a cross table** means going beyond a simple two-variable table and examining the relationship in **greater detail**, often by introducing another variable.

Let's first consider a simple table.

Gender × Employment

Gender	Employed	Unemployed
Male	40	10
Female	25	25

We might conclude that males have a higher employment rate.

But now suppose we introduce **education** as a third variable.

Gender × Employment, according to Education

Continue....

47

- Graduates

Gender	Employed	Unemployed
Male	30	5
Female	25	5

- Non-graduates

Gender	Employed	Unemployed
Male	10	5
Female	0	20

Elaboration means breaking down or expanding the original cross-tabulation to understand the relationship more deeply.

Now we have a more detailed picture.

It may be that the apparent relationship between **gender and employment** is partly associated with **education**.

5. Spearman's Rank Order Correlation Coefficient

48

This is different from cross-tabulation.

- Spearman's rank-order correlation coefficient** is used to measure the **direction and strength of association between two variables when the data can be ranked**.

It is commonly represented by:

r_s

Its value ranges from:

-1 to +1

Meaning of the values

Value of r_s	Interpretation
+1	Perfect positive association
Close to +1	Strong positive association
0	No rank association
Close to -1	Strong negative association
-1	Perfect negative association

49

Formula

When there are **no tied ranks**, the formula is:

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Where:

- r_s = Spearman's rank correlation coefficient
- d = difference between the two ranks
- d^2 = square of the difference
- n = number of observations
- $\sum d^2$ = sum of all squared rank differences

Example

50

- Suppose a researcher wants to examine whether **study time** is related to **examination performance** for five students.

Student	Study Time Rank (X)	Performance Rank (Y)	$d = X - Y$	d^2
A	1	2	-1	1
B	2	1	1	1
C	3	3	0	0
D	4	5	-1	1
E	5	4	1	1
Total				4

51

Continue....

Therefore:

$$\sum d^2 = 4$$

and

$$n = 5$$

Now apply the formula:

$$\begin{aligned} r_s &= 1 - \frac{6(4)}{5(5^2 - 1)} \\ &= 1 - \frac{24}{5(24)} \\ &= 1 - \frac{24}{120} \\ r_s &= 0.80 \end{aligned}$$

Interpretation

The value is:

$$r_s = +0.80$$

This indicates a strong positive relationship between study-time rank and examination-performance rank.

Students who rank higher in study time generally tend to rank higher in examination performance.

52

What is rank order and data transformation in data analysis?

53

Analysis of Data

After collecting data, the researcher has to **organize, classify, and transform the data** so that meaningful conclusions can be drawn.

Two useful procedures are:

1. Calculating Rank Order
2. Data Transformation

54

Calculating Rank Order

- **Rank order means arranging observations from highest to lowest or lowest to highest according to a particular characteristic.**
- The first position is given **Rank 1**, the second position **Rank 2**, and so on.
- Rank ordering is useful when the researcher wants to know the **relative position** of different individuals, groups, products, preferences, etc.

55

Example:

- Suppose five students obtain the following marks:

Student	Marks	Rank
A	85	1
B	72	3
C	90	1*
D	65	5
E	78	2

There is actually a problem here: A and C cannot both simply be Rank 1 because C has the highest mark. Correct ranking would be:

Student	Marks	Rank
C	90	1
A	85	2
E	78	3
B	72	4
D	65	5

Thus, rank order converts the original numerical values into their relative positions.

56

Data Transformation

- **Data transformation means changing the original form of data into another form that is more suitable for analysis, comparison, or interpretation.**
- The original information is not necessarily changed in meaning. Rather, it is **re-expressed or recoded** into a form that makes analysis easier.
- For example, a researcher may transform:
 - ❑ raw scores into percentages;
 - ❑ numerical ages into age groups;
 - ❑ income values into income categories;
 - ❑ responses into codes;
 - ❑ frequencies into percentages;
 - ❑ continuous data into categories.

Example

57

Suppose the researcher collects the following ages:

18, 21, 23, 27, 31, 35, 42, 48

The researcher may transform these raw ages into age groups:

Age	Age group
18	18–25
21	18–25
23	18–25
27	26–35
31	26–35
35	26–35
42	36–45
48	46–55

Now it becomes easier to compare respondents by **age category**.

Example: Data transformation into percentages

58

Suppose a survey of 100 respondents gives:

Response	Frequency
Yes	70
No	30

The researcher can transform these frequencies into percentages:

$$\text{Percentage} = \frac{\text{Frequency}}{\text{Total}} \times 100$$

Therefore:

$$\text{Yes} = \frac{70}{100} \times 100 = 70\%$$

$$\text{No} = \frac{30}{100} \times 100 = 30\%$$

The transformed table becomes:

Response	Percentage
Yes	70%
No	30%

This makes interpretation easier.

59

Thanks